

⇒ Derivative of Laplace:

→ solve the homogeneous diff eq by Laplace

→ Given that:

$$y''' + 2y'' - y' - 2y = 0$$

$$\begin{aligned} y(0) &= y'(0) = 0 \\ y''(0) &= 6 \end{aligned}$$

General form:

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{f(s)\}$$

$$\mathcal{L}\{f'(t)\} = \mathcal{L}[sf(s) - f(0)]$$

$$\mathcal{L}\{f''(t)\} = [s^2 f(s) - sf(0) - f'(0)]$$

$$\mathcal{L}\{f'''(t)\} = [s^3 f(s) - s^2 f(0) - sf'(0) - f''(0)]$$

so now

Applying Laplace on both sides

$$\mathcal{L}\{y'''\} + 2\mathcal{L}\{y''\} - \mathcal{L}\{y'\} - 2\mathcal{L}\{y\} = \mathcal{L}\{0\}$$

$$[s^3 f(s) - s^2 f(0) - sf'(0) - f''(0)] + 2[s^2 f(s) - sf(0) - f'(0)] - [sf(s) - f(0)] - 2sf(s) = 0$$

we know that,

$$y(0) = y'(0) = 0$$

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Applying Laplace on both sides

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$$\begin{aligned} &[s^3 f(s) - s^2 f(0) - sf'(0) - f''(0)] + 2[s^2 f(s) - sf(0) - f'(0)] \\ &- [sf(s) - f(0)] - 2sf(s) = 0 \end{aligned}$$

we know that,

$$y(0) = y'(0) = 0$$

$$\boxed{y''(0) = 6}$$

$$s^3 f(s) - f''(0) + 2s^2 f(s) - s f(s) - 2s f(s) = 0$$

$$s^3 f(s) - 6 + 2s^2 f(s) - 2s f(s) = 0$$

$$f(s) [s^3 + 2s^2 - 2s - 2] = 6$$

$$f(s) = \frac{6}{s^3 + 2s^2 - s - 2} \rightarrow f(s) \quad \text{Now by using inverse of Laplace}$$

$$\rightarrow s^3 + 2s^2 - s - 2$$

$$[n=1]$$

$$1 - 2 - 1 - 2 = 0$$

$$[s = -1]$$

$$-1 - 2 + 1 - 2$$

$$(-1)^3 - 2(-1)^2 - (-1) - 2$$

$$-1 - 2 + 1 - 2$$

$$[s = -1] \checkmark$$

Now

-1	1	2	-1	-2
	+	-1	-1	2
	1	-1	-2	0

$$x^2 + 2x - 2 = 0$$

$$x^2 + x - 2 = 0$$

$$x^2 + 2x + x - 2$$

$$x^2 - x + 2x - 2 = 0$$

$$x(x-1) + 2(x-1)$$

$$(x+2)(x-1) = 0$$

$$[x = -2] \quad [x = 1]$$

So roots are real and distinct

$$f(s) = \frac{c_1 e^{s_1 t} + c_2 e^{s_2 t}}{s}$$

but

we have to find inverse so

6

$$(x+1)(x+2)(x-1)$$

$$\frac{6}{(n+1)(n+2)(n-1)} = \frac{A}{(n+1)} + \frac{B}{(n+2)} + \frac{C}{(n-1)}$$

$$\frac{6}{(n+1)(n+2)(n-1)} = \frac{A}{(n-1)} + \frac{B}{(n+1)} + \frac{C}{(n+2)}$$

$$6 = A(n+1)(n+2) + B(n-1)(n+2) + C(n-1)(n+1)$$

$$\boxed{n = -1}$$

$$6 = A(-1+1)(-1+2) + B(-1-1)(-1+2) + C(-1-1)(-1+1)$$

$$6 = 0 - 2B + 0$$

$$6 = -2B$$

$$\frac{6}{-2} = B$$

$$\boxed{-3 = B}$$

$$B = -3 \checkmark$$

$$\text{Now } \boxed{n = 1}$$

$$6 \neq A(1+1)(1+2) + B(1-1)(1+2) + C(1-1)(1+1)$$

$$6 = 6A + 0 + 0$$

$$\boxed{A = 1} \checkmark$$

$$\boxed{n = -2}$$

$$6 = A(-2+1)(-2+2) + B(-2-1)(-2+2) + C(-2-1)(-2+1)$$

$$6 = 0 + 0 + 3C$$

$$\frac{6}{3} = C$$

$$\boxed{C = 2}$$

$$\frac{6}{(n+1)(n+2)(n-1)} = \frac{+1}{n-1} + \frac{3}{(n+1)} + \frac{2}{(n+2)}$$

$$+ \left\{ \frac{1}{(s-1)} \right\} - \left\{ \frac{3}{(s+1)} \right\} + 2 \left\{ \frac{1}{s+2} \right\}$$

$$\boxed{+ e^t - 3e^{-t} + 2e^{-2t}}$$

Ans.

Q5

$$y'' - 4y' + 4y = (12x^2 - 6x)e^{2x}$$

Given $y(0) = 1$ and $y'(0) = 0$

$$\{y''\} - 4\{y'\} + 4\{y\} = \{12x^2 - 6x\} \{e^{2x}\}$$

$$[s^2 f(s) - s f(0) - f'(0)] - 4[s f(s) - f(0)] + 4[f(s)] =$$

$$s^2 f(s) - \frac{s f(0)}{1} - \frac{f'(0)}{0} - 4s f(s) + 4 \frac{f(0)}{1} + 4 f(s) =$$

$$s^2 f(s) - 1 - 0 - 4s f(s) + 4(1) + 4 f(s) =$$

$$[s^2 f(s) - s - 4s f(s) + 4 f(s)] + 4 =$$

$$[s^2 f(s) - 4s f(s) + 4 f(s)] 4 - s =$$

$$f(s) [s^2 - 4s + 4] = (s-4) + \left[12 \frac{2!}{s^{2+1}} - 6 \frac{1!}{s^{1+1}} \right] e^{2x}$$

$$f(s) [s^2 - 4s + 4] = (s-4) + \left[\frac{24}{s^3} - \frac{6}{s^2} \right] e^{2x}$$

$$f(s) (s-2)^2 = s-4 + \frac{24}{(s-2)^3} - \frac{6}{(s-2)^2}$$

$$f(s) = \frac{(s-4)}{(s-2)^2} + \frac{24}{(s-2)^3} - \frac{6}{(s-2)^4}$$

→ Now find the inverse

$$f(s) = \left(\frac{s-4}{(s-2)^2} \right) + \left(\frac{24}{(s-2)^5} \right) - \left(\frac{6}{(s-2)^4} \right) \quad ???$$

$$ne^{2n} + 24 \cdot \frac{n^4 e^{2n}}{4!} - 6 \cdot \frac{n^3 e^{2n}}{3!} = \frac{2^3 2n}{3!} = -n^3 e^{2n}$$

$$f(s) = e^{2n} (n + n^4 - n^3)$$

→ Now prove

$$ne^{2n} = \frac{1!}{s^{1+1}} = \frac{1}{s^2} e^{2n} = \frac{1}{(s-2)^2}$$

$$24 \times \frac{1}{(s-2)^5} = 24 \frac{n^4}{4!} e^{2n} = 24 \frac{n^4}{4 \times 3 \times 2 \times 1} e^{2n}$$

$$\rightarrow \mathcal{L}[n] = \frac{1}{s^2}, \text{ the } \mathcal{L}[e^{2n} n] = \frac{1}{(s-2)^2}$$